

Responses to the Review Comments of BDR-D-25-00303R3

Original Manuscript ID: BDR-D-25-00303R3

Original Article Title: “Dual Balanced Large Marginal Machine for Imbalanced Data Classification”

To: Big Data Research Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, and we particularly appreciate the Editor-in-Chief for granting us the opportunity to conduct a round of revisions to address the reviewers' comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with orange highlighting indicating changes, and (c) an Supplementary file submitted in accordance with the reviewers' requirements.

Best wishes,

Pingping Gu, Denghao Dong, Quan Shi, Fudi Wang and Guojun Huang*

The response letter is structured as follows:

- Response for Editor: Page 2.
- Response for Reviewer 7: Page 3.
- Response for Reviewer 8: Page 4-8.

EDITOR

General Comments:

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following minor revision and modification. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by 07/05/2026. When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

Response: We sincerely appreciate the Editor-in-Chief and all reviewers for their constructive feedback and for granting us the opportunity to further refine our manuscript. We have carefully addressed all comments raised in this fourth round of review. The detailed point-by-point responses will be added below under Sections “Reviewer 7” and “Reviewer 8”.

The point-by-point responses to Reviewer #7 and Reviewer #8 will be added in the sections below. Author responses and author actions are left blank for now and will be completed during the revision. To facilitate reviewers in checking our revisions, we have prepared detailed response documents. The concrete revisions and responses to the reviewers are presented in Sections “Reviewer 7” and “Reviewer 8”. Note that reference numbers are in accordance with the references at the end of **this response document**. For clarity, we present the section or subsection, page, and line numbers of the revisions based on **the newly revised version manuscript**, NOT the previous manuscript. Moreover, we mark the revised contents in the manuscript using **magenta color** to distinguish them from previous revisions. The detailed revisions are presented as follows.

REVIEWER 7

General Comments: The manuscript has undergone three rounds of revision and is now highly polished. The authors have addressed the reviewers' concerns regarding comparison with the latest state-of-the-art models, scalability for Big Data paradigms, experimental reproducibility, and terminology consistency. The paper provides a meaningful incremental advance in margin distribution learning. While the $O(n^3)$ complexity is a hurdle for "Big Data," the provided theoretical paths toward distributed and approximated versions justify its publication.

Reviewer 7, Final Suggestion: Before final publication, ensure that the "Supplementary Material" links are finalized and that all magenta/orange highlighting from the revision process is removed for the final production version.

REVIEWER 8

General Comments: The authors have moved the manuscript forward in this round. The new Remark 2 clarifying “quantile distance” versus “composite distance” is useful, the MLP+Focal configuration is now properly specified, and the limitations discussion in the conclusion is a real improvement. That said, the new head-to-head against DeepSMOTE, LDAM, and Balanced Softmax introduces problems that need to be resolved, and several issues that should have been caught in earlier rounds are still present. I recommend major revision.

Reviewer 8, Strength 1: The new Remark 2 clearly distinguishes the two distance concepts that were previously conflated. This was a real source of confusion in R2 and the fix is well done.

Reviewer 8, Strength 2: The expanded MLP+Focal configuration in Section 5.3 (architecture, optimizer, β values, batch size, LR schedule, grid search ranges) is detailed enough for reproducibility.

Reviewer 8, Strength 3: The limitations paragraph in the conclusion (training complexity, extreme imbalance, concept drift, unstructured data) gives a more balanced positioning of the method than earlier versions.

Reviewer 8, Strength 4: The scalability discussion (ADMM, mini-batch, Random Fourier Features) at least acknowledges the journal’s scope, even if it is not yet demonstrated.

Reviewer 8, Major Concern 1: The LDAM configuration as described is methodologically incoherent. The authors write that LDAM “adapts ResNet-18 for tabular data.” ResNet-18 is a 2D convolutional architecture designed for image inputs. There is no standard way to apply it to tabular features of dimension 3 to 61, which is the range in Table 1. The authors need to explain exactly what architecture was used, replace it with a sensible tabular backbone (MLP with LDAM loss would be the natural choice), or drop the claim. As it stands, readers cannot reproduce or evaluate this baseline.

Reviewer 8, Major Concern 2: Please release the code, seeds, and fold-level outputs for Table 7. The ordering LDAM over Balanced Softmax over DeepSMOTE is very consistent across the 17 datasets and across Accuracy, AUC, and F1. This ordering is plausible on its own, but the consistency is unusual enough that the paper should provide a reproducibility package (code, random seeds, per-fold numbers). This would also address Reviewer 5’s original reproducibility request in a more complete way.

Reviewer 8, Major Concern 3: The Wilcoxon p-value reporting needs more detail. Reporting $p = 0.03125$ for all four comparisons (DBUPLDM vs SVM, UPSVM, PinSVM, LDM) is consistent with a test over 5 or 6 paired observations where DBUPLDM wins all of them, but the paper does not say what the paired units are (datasets? η values? $\eta \times$ datasets?) or what n is. Please state the test setup and report effect sizes alongside the p-values. A one-line description would clear this up.

Reviewer 8, Major Concern 4: Text-table inconsistency. Section 5.3 says “deep learning methods show strong performance on certain datasets (e.g., LDAM on Ionosphere and Australia).” In Table 7, DBUPLDM beats LDAM on Australian (87.24 vs 84.20). Pick examples that actually match the table, or remove the specific claim.

Reviewer 8, Major Concern 5: Big Data Research scope is still only rhetorical. The new scalability paragraph in the conclusion discusses ADMM, mini-batch processing, and Random Fourier Features, but none of this is implemented or evaluated. The experiments are entirely on small UCI datasets (hundreds to about a thousand samples). At minimum, I would ask for one empirical demonstration on a larger-scale imbalanced dataset (for example, CreditCard Fraud with about 285k samples, KDD99, or a similar public benchmark) to support the scalability claims. Without this, the journal fit remains weak.

Reviewer 8, Major Concern 6: Novelty over the authors’ own prior work should be spelled out. The UP loss applied to a large-margin distribution classifier already appears in Gu et al. 2024 (cited in the paper). The dual balance factor is close in spirit to Cheng et al.’s CS-LDM (2016). The paper’s contribution is essentially the combination of these two ingredients. A short paragraph in the introduction or Section 3 stating what is new here relative to Gu et al. 2024 would strengthen the case.

Reviewer 8, Minor Issue 1: Title: “Dual Balanced Large Marginal Machine” should be “Large Margin Machine” (or “Large Margin Distribution Machine” to match the abstract). This has remained wrong across rounds and needs to be fixed.

Reviewer 8, Minor Issue 2: Dataset naming is inconsistent. Table 1 lists “Herberman,” the results tables use “Heberman,” and Table 7 uses “Habe.” The dataset is Haberman’s Survival. Pick one spelling and use it throughout.

Reviewer 8, Minor Issue 3: “Ecoil” should be “Ecoli” (E. coli dataset).

Reviewer 8, Minor Issue 4: “Germans” should likely be “German” (German Credit dataset).

Reviewer 8, Minor Issue 5: “Phonemes” vs “Phoneme” is inconsistent between Table 1 and Section 5.5.

Reviewer 8, Minor Issue 6: Table 7, WDBC row: “87.50/91.43/0/89” has a slash where a decimal should be. Should read “0.89.”

Reviewer 8, Minor Issue 7: Figure 6, subfigures (i), (j), (k), and (l) are all captioned “Prlx dataset.” These are clearly different datasets. Fix the captions.

Reviewer 8, Minor Issue 8: Figure 6(g) caption reads “Daibetes dataset.” Should be “Diabetes.”

Reviewer 8, Minor Issue 9: Section 5.5: “PinSVM, UPSVM, and DBUPLDM) performe well” should be “perform.”

Reviewer 8, Minor Issue 10: The Section 1 roadmap reads “Section 3 proposes the Dual Balanced method and DBUPLDM model. Section 5 discusses...” Section 4 (Properties of DBUPLDM) is skipped. Add it.

Reviewer 8, Minor Issue 11: Table 5 caption “Minimum distance with second order statistics” does not describe the content (ablation over balanced factors and hinge vs UP loss). Rewrite.

Reviewer 8, Minor Issue 12: Remark renumbering: the new Remark 2 (Terminology Clarification) shifts former Remark 2 to Remark 3 and former Remark 3 to Remark 4. Please verify no downstream cross-references still point to the old numbers.

Reviewer 8, Minor Issue 13: The third contribution bullet in the introduction says “the quantile distance is introduced to replace the shortest distance between sample sets in traditional LDM.” Given the new Remark 2, this is imprecise (the replacement involves both the UP loss and the composite distance). Reword for consistency.

Reviewer 8, Minor Issue 14: The response letter header says “BDR-D-25-00303R3” but the opening paragraph says “R1,” and later describes this as the third round. Fix this.

Reviewer 8, Minor Issue 15: Reviewer 6’s comment about “deeper theoretical insight” is acknowledged in the response but not concretely addressed. A brief sentence on what was done, or why no change was needed, would help.

ACKNOWLEDGEMENT

Special thanks to you for the valuable and helpful comments and suggestions from editors and anonymous reviewers. The comments are helpful in improving our paper, as well as the important guiding significance to our research. We hope that the editors' and reviewers' concerns have been adequately addressed in the revised version.

Yours sincerely,

Pingping Gu, Denghao Dong, Quan Shi, Fudi Wang and Guojun Huang*